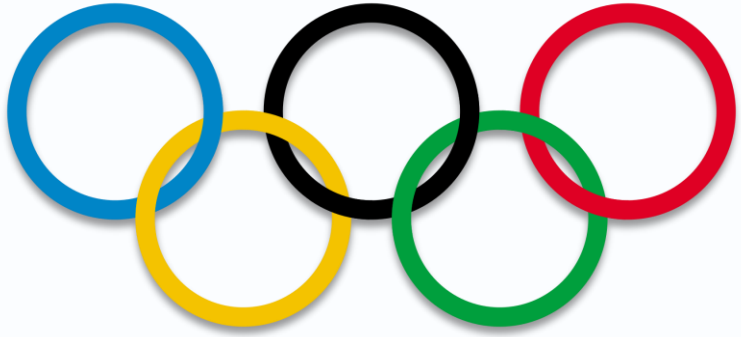


# Olympic Power BI Project

## Olympic Insights Using Data Analytics

Presented by Kunal Malviya

# Introduction

## 1 The Problem

The IOC wants deeper insight into athlete performance, country-wise medal trends and sport-specific dominance across the Olympic Games.

## 2 The Data

Historical Olympic data modelled in Power BI: athletes, event results, medal tally, host cities and editions, linked through relationships.

## 3 The Analysis

32 KPI questions solved with DAX measures across six sections, from global medals to gender and age demographics.

## 4 The Dashboard

A six-page navigational Power BI dashboard with buttons to track achievements and support strategic planning.

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- Year-wise Total Events Held

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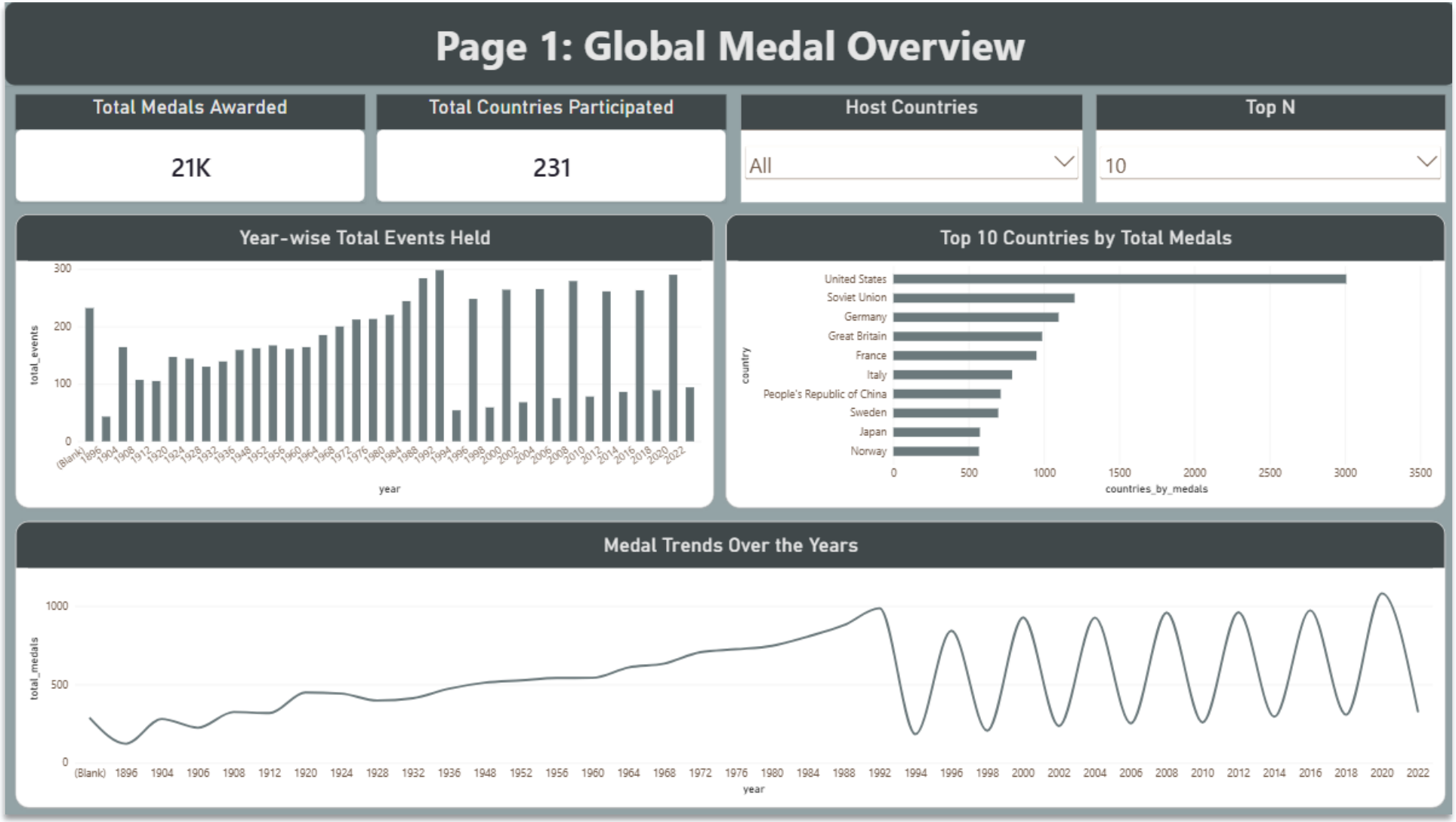
- Total Olympics Hosted per Country
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- Gender-wise Medal Distribution
- Youngest & Oldest Medal Winners
- Avg. Age of Participants by Year
- Participation by Age Group

SECTION 1

# 1 Global Medal Overview



# Q1 Total Medals Awarded

```
1 total_medals =
2       SUMX(
3           olympic_games_medal_tally,
4           olympic_games_medal_tally[total]
5           )
```

| Total Medals Awarded |
| --- |
| 21K |

# Q2 Total Countries Participated

```
1 total_countries_participated =
2     DISTINCTCOUNT(
3         olympic_athlete_event_results[country_noc]
4         )
```

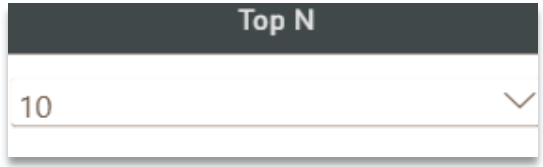
| Total Countries Participated |
| --- |
| 231 |

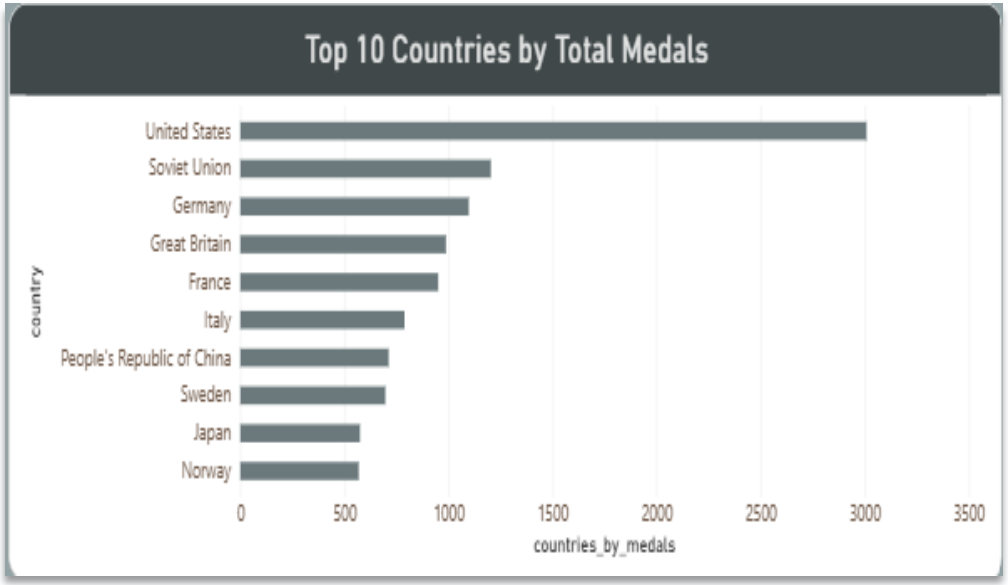
# Q3 Top 10 Countries by Total Medals

```
countries_by_medals =

VAR ranking =
    RANKX(
        ALLSELECTED(
            olympic_games_medal_tally[country]
            ),
        [total_medals],,DESC,Dense
    )

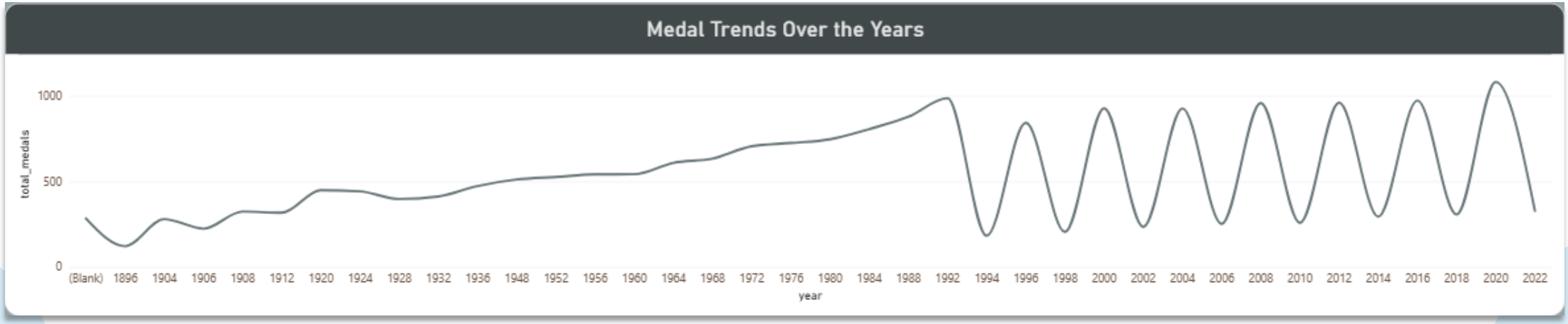
 RETURN
    IF(
        ranking <= Parameter[Parameter Value],
        [total_medals])
```



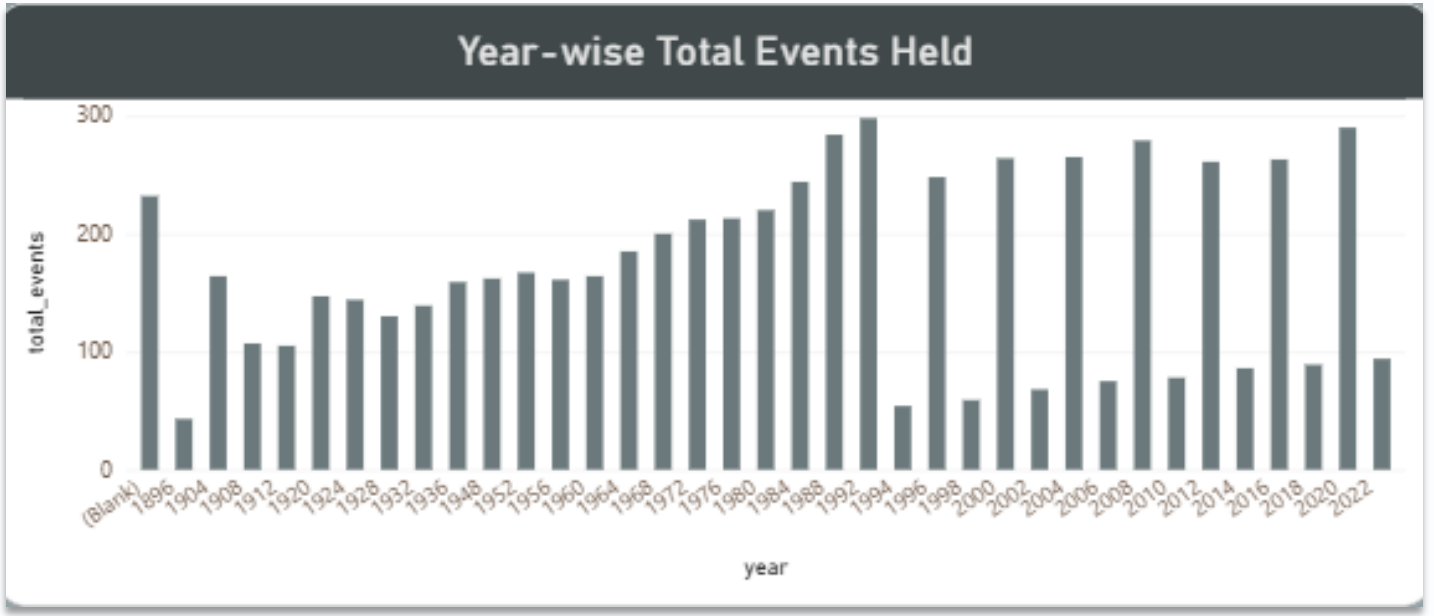
# Q4 Medal Trends Over the Years

```
total_medals =
    SUMX(
        olympic_games_medal_tally,
        olympic_games_medal_tally[total]
        )
```



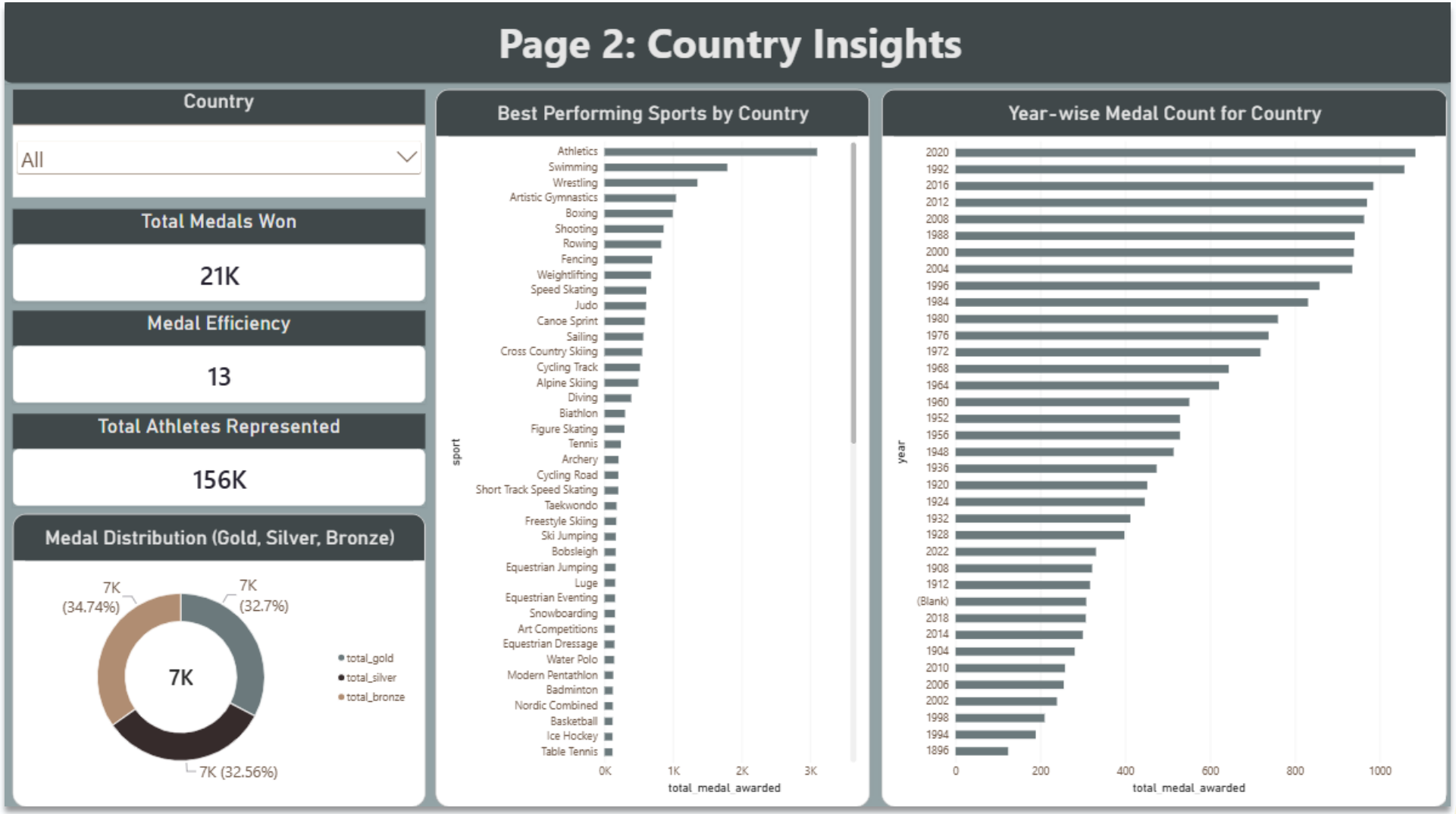
# Q5 Year-wise Total Events Held

```
1 total_events =
2     DISTINCTCOUNT(
3         olympic_athlete_event_results[event]
4         )
```



SECTION 2

# 2 Country Insights



# Q1 Total Athletes Represented

```
athlete_represented =
    DISTINCTCOUNT(
        olympic_athlete_event_results[athlete_id]
        )
```

| Total Athletes Represented |
| --- |
| 156K |

# Q2 Total Medals Won

```
total_medal_awarded =

VAR IndividualMedals =
    CALCULATE(
        COUNTROWS(olympic_athlete_event_results),
        olympic_athlete_event_results[isTeamSport] = "False",
        olympic_athlete_event_results[medal] <> ""
    )

VAR TeamMedals =
    COUNTROWS(
        CALCULATETABLE(
            SUMMARIZE(
                olympic_athlete_event_results,
                olympic_athlete_event_results[result_id],
                olympic_athlete_bio[country_noc],
                olympic_athlete_event_results[medal]
            ),
            olympic_athlete_event_results[isTeamSport] = "True",
            olympic_athlete_event_results[medal] <> ""
        )
    )

RETURN
    IndividualMedals + TeamMedals
```

| Total Medals Won |
| --- |
| 21K |

# Q3 Medal Efficiency (Medals per 100 Athletes)

```
1 medal_efficiency =
2     DIVIDE(
3         [total_medal_awarded],
4         [athlete_represented],
5         0
6         ) * 100
```

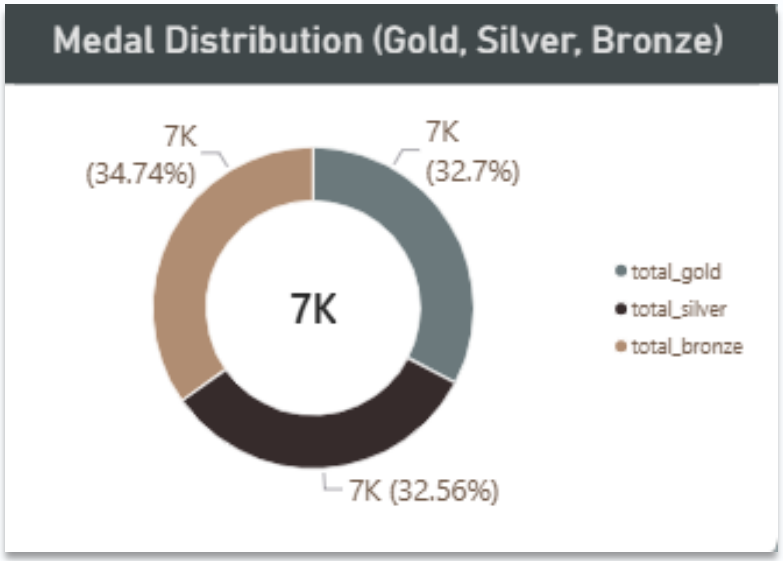
| Medal Efficiency |
| --- |
| 13 |

# Q4 Medal Distribution (Gold, Silver, Bronze)

```
total_bronze =
    SUMX(
        olympic_games_medal_tally,
        olympic_games_medal_tally[bronze]
    )
```

```
total_silver =
    SUMX(
        olympic_games_medal_tally,
        olympic_games_medal_tally[silver]
    )
```

```
total_gold =
    SUMX(
        olympic_games_medal_tally,
        olympic_games_medal_tally[gold]
    )
```



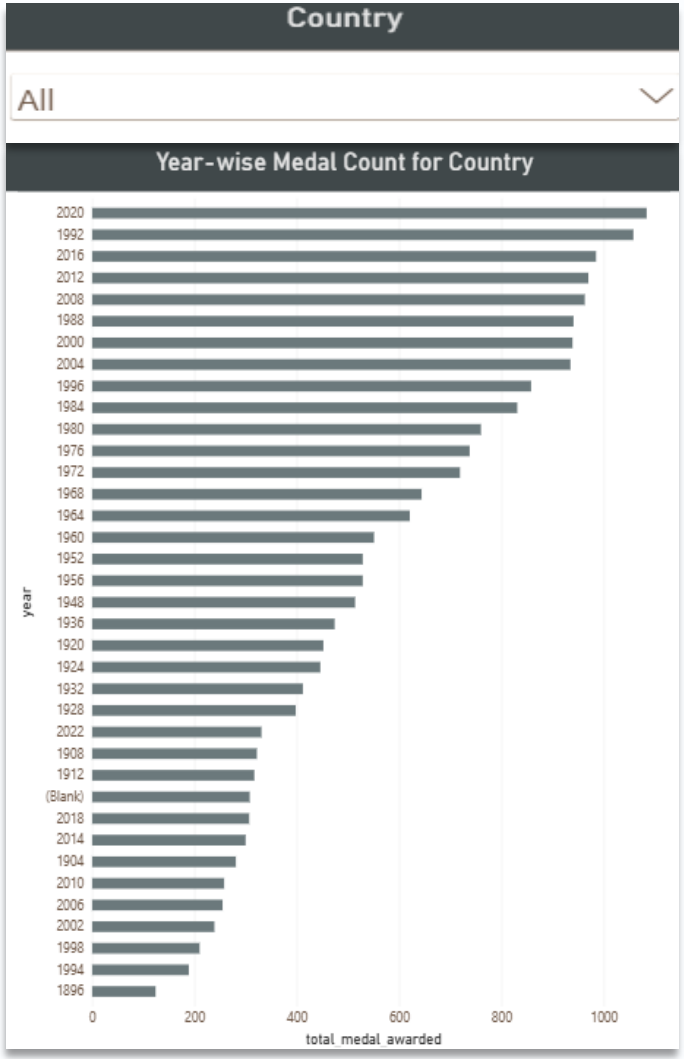
# Q5 Year-wise Medal Count for Country

```
total_medal_awarded =

VAR IndividualMedals =
    CALCULATE(
        COUNTROWS(olympic_athlete_event_results),
        olympic_athlete_event_results[isTeamSport] = "False",
        olympic_athlete_event_results[medal] <> ""
    )

VAR TeamMedals =
    COUNTROWS(
        CALCULATETABLE(
            SUMMARIZE(
                olympic_athlete_event_results,
                olympic_athlete_event_results[result_id],
                olympic_athlete_bio[country_noc],
                olympic_athlete_event_results[medal]
            ),
            olympic_athlete_event_results[isTeamSport] = "True",
            olympic_athlete_event_results[medal] <> ""
        )
    )

RETURN
    IndividualMedals + TeamMedals
```



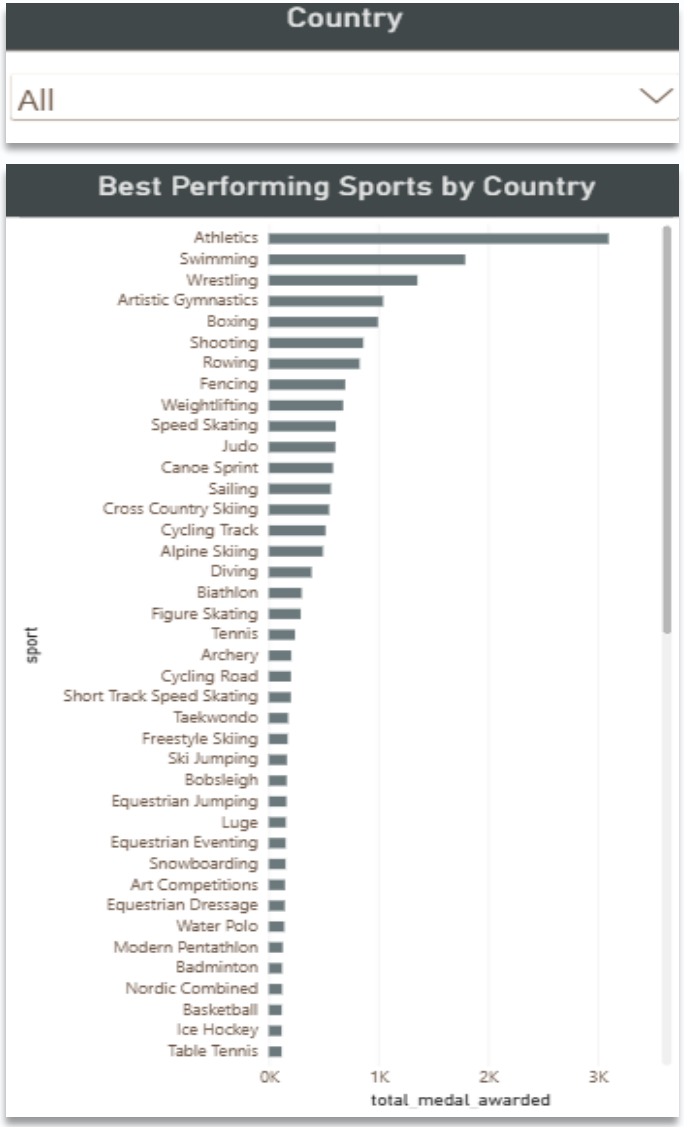
# Q6 Best Performing Sports by Country

```
total_medal_awarded =

VAR IndividualMedals =
    CALCULATE(
        COUNTROWS(olympic_athlete_event_results),
        olympic_athlete_event_results[isTeamSport] = "False",
        olympic_athlete_event_results[medal] <> ""
    )

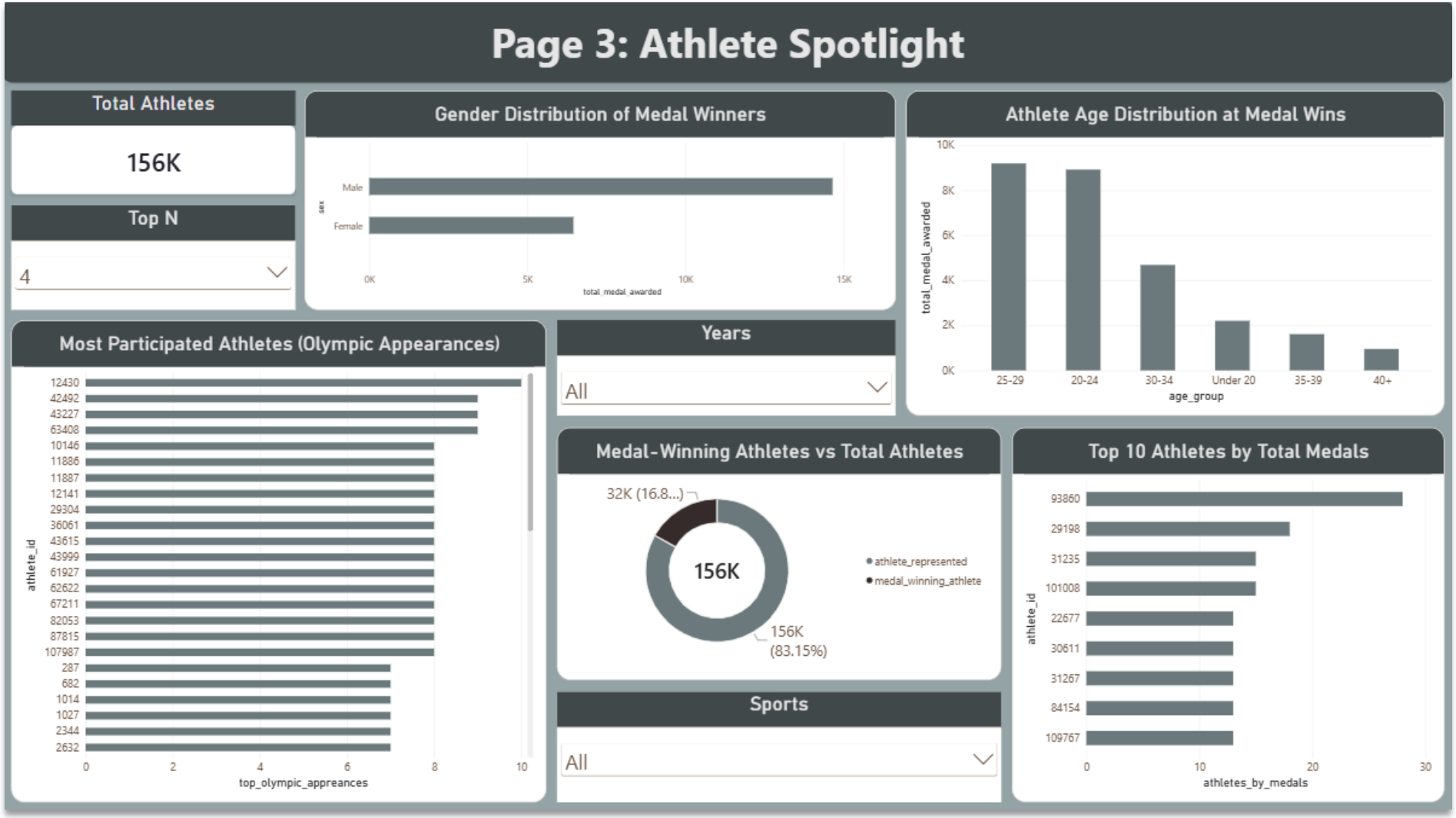
VAR TeamMedals =
    COUNTROWS(
        CALCULATETABLE(
            SUMMARIZE(
                olympic_athlete_event_results,
                olympic_athlete_event_results[result_id],
                olympic_athlete_bio[country_noc],
                olympic_athlete_event_results[medal]
            ),
            olympic_athlete_event_results[isTeamSport] = "True",
            olympic_athlete_event_results[medal] <> ""
        )
    )

RETURN
    IndividualMedals + TeamMedals
```



SECTION 3

# 3 Athlete Spotlight



# Q1 Total Athletes

```
1 athlete_represented =
2     DISTINCTCOUNT(
3         olympic_athlete_event_results[athlete_id]
4         )
```

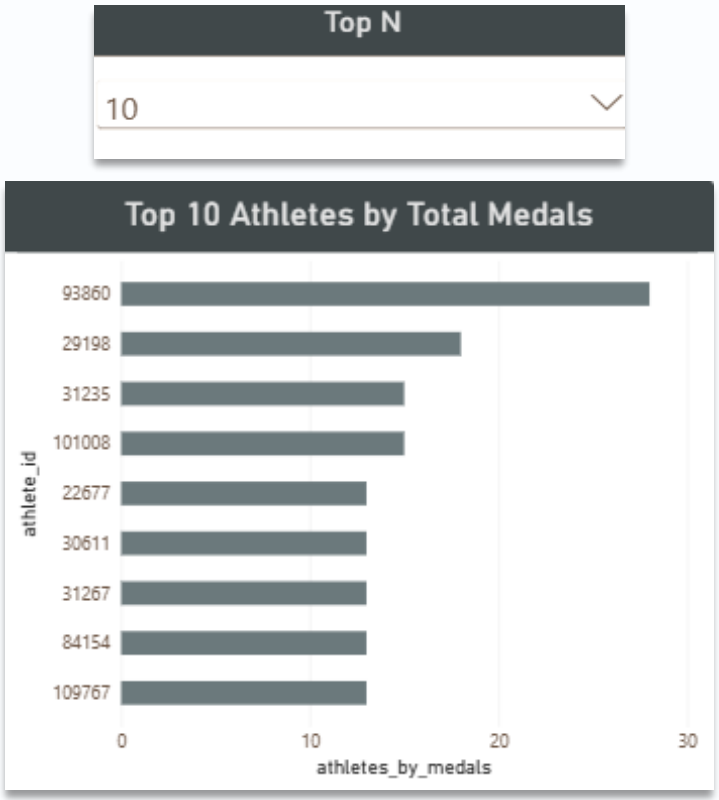
| Total Athletes |
| --- |
| 156K |

# Q2 Top 10 Athletes by Total Medals

```
athletes_by_medals =

VAR ranking =
    RANKX(
        ALLSELECTED(
            olympic_athlete_bio[athlete_id]
            ),
            [total_medal_awarded], , DESC, Dense
    )

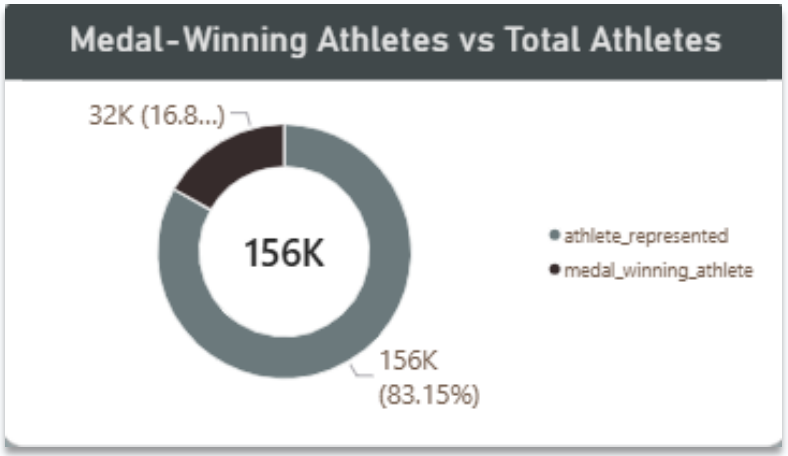
RETURN
    IF(
        ranking <= Parameter[Parameter Value],
        [total_medal_awarded]
    )
```



# Q3 Medal-Winning Athletes vs Total Athletes

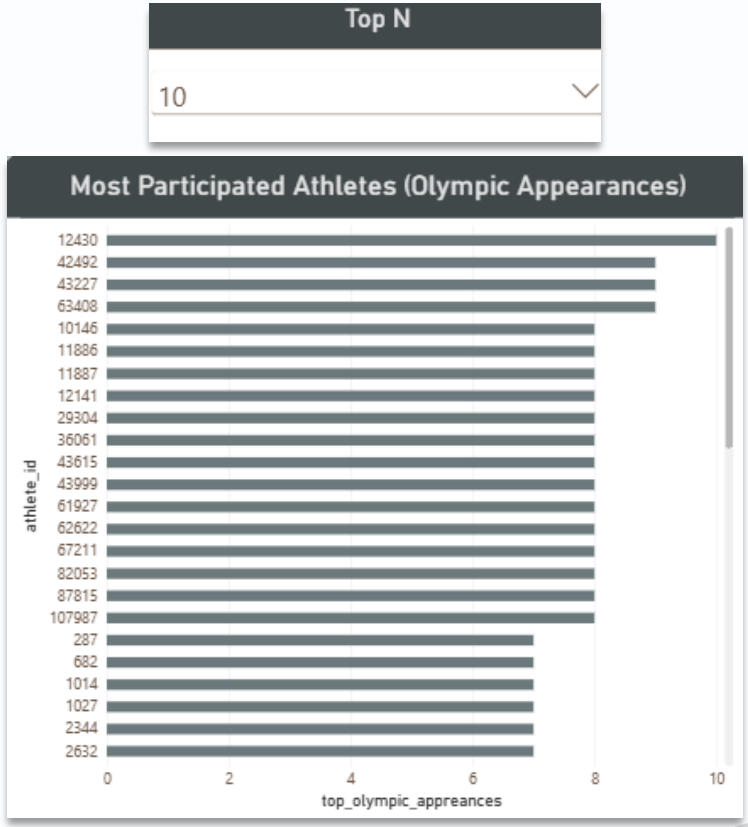
```
medal_winning_athlete =
    CALCULATE(
        DISTINCTCOUNT(
            olympic_athlete_event_results[athlete_id]),
            olympic_athlete_event_results[medal] IN {"Gold", "Silver", "Bronze"}
    )
```

```
athlete_represented =
    DISTINCTCOUNT(
        olympic_athlete_event_results[athlete_id]
        )
```



# Q4 Most Participated Athletes (Olympic Appearances)

```
1 top_olympic_appreances =
2
3 VAR ranking =
4     RANKX(
5         ALLSELECTED(olympic_athlete_event_results[athlete_id]),
6         [olympic_appreances],, DESC, Dense
7     )
8
9 RETURN
10     IF(ranking <= Parameter[Parameter Value],
11     [olympic_appreances]
12     )
```



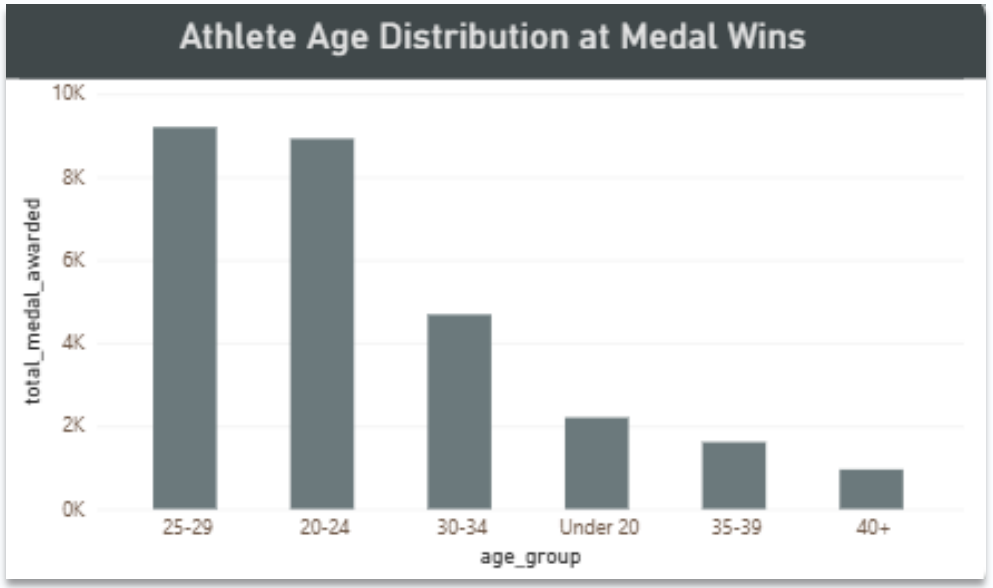
# Q5 Athlete Age Distribution at Medal Wins

```
total_medal_awarded =

VAR IndividualMedals =
    CALCULATE(
        COUNTROWS(olympic_athlete_event_results),
        olympic_athlete_event_results[isTeamSport] = "False",
        olympic_athlete_event_results[medal] <> ""
    )

VAR TeamMedals =
    COUNTROWS(
        CALCULATETABLE(
            SUMMARIZE(
                olympic_athlete_event_results,
                olympic_athlete_event_results[result_id],
                olympic_athlete_bio[country_noc],
                olympic_athlete_event_results[medal]
            ),
            olympic_athlete_event_results[isTeamSport] = "True",
            olympic_athlete_event_results[medal] <> ""
        )
    )

RETURN
    IndividualMedals + TeamMedals
```



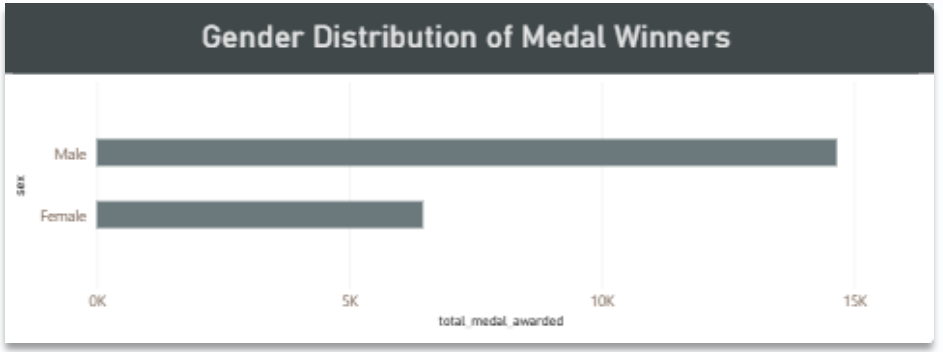
# Q6 Gender Distribution of Medal Winners

```
total_medal_awarded =

VAR IndividualMedals =
    CALCULATE(
        COUNTROWS(olympic_athlete_event_results),
        olympic_athlete_event_results[isTeamSport] = "False",
        olympic_athlete_event_results[medal] <> ""
    )

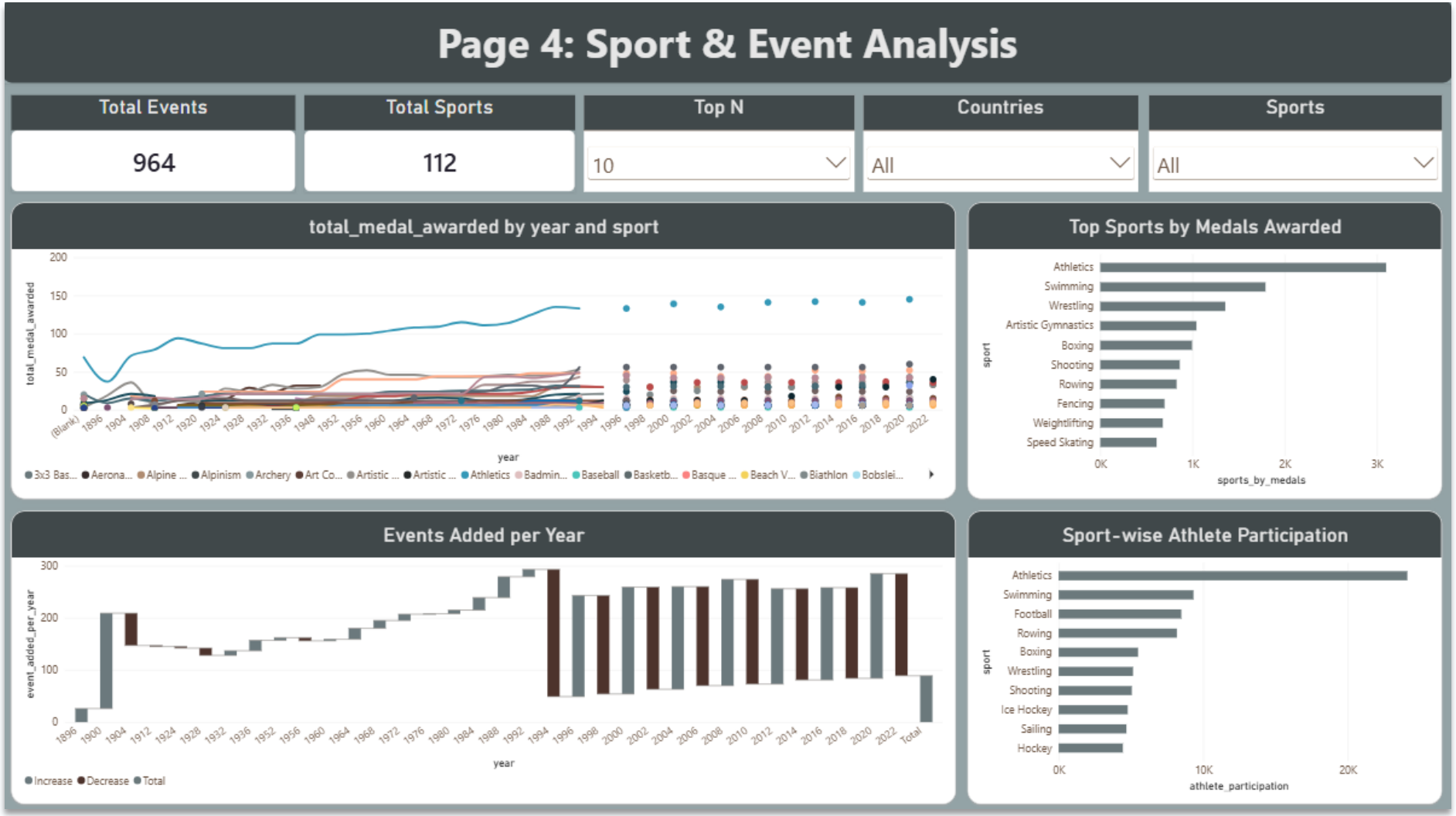
VAR TeamMedals =
    COUNTROWS(
        CALCULATETABLE(
            SUMMARIZE(
                olympic_athlete_event_results,
                olympic_athlete_event_results[result_id],
                olympic_athlete_bio[country_noc],
                olympic_athlete_event_results[medal]
            ),
            olympic_athlete_event_results[isTeamSport] = "True",
            olympic_athlete_event_results[medal] <> ""
        )
    )

RETURN
    IndividualMedals + TeamMedals
```



SECTION 4

# 4 Sport & Event Analysis



# Q1 Total Sports/Events

```
1 total_sports =
2     DISTINCTCOUNT(
3         olympic_athlete_event_results[sport]
4     )
```

| Total Sports |
| --- |
| 112 |

```
1 total_events =
2     DISTINCTCOUNT(
3         olympic_athlete_event_results[event]
4         )
```

| Total Events |
| --- |
| 964 |

# Q2 Events Added per Year

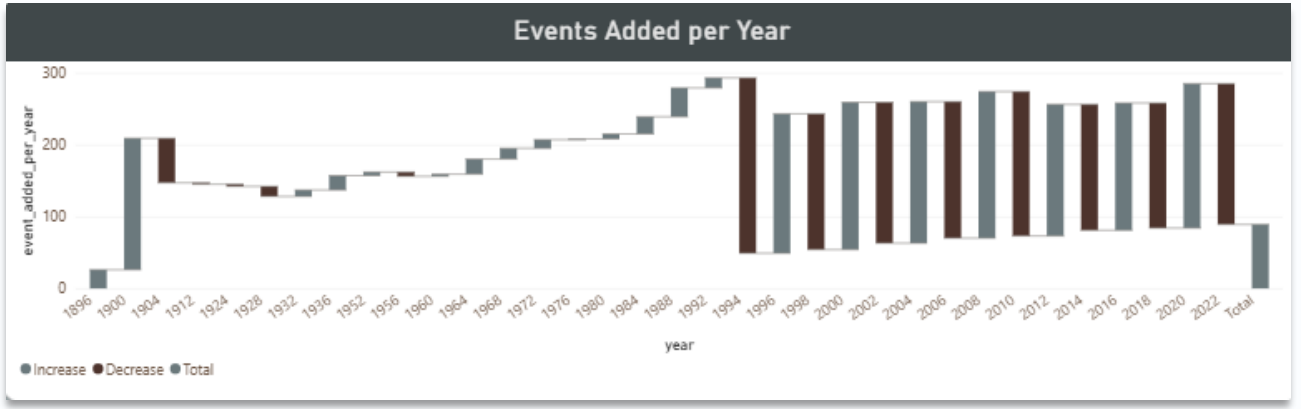
```
prev_year_events =

VAR curr_year =
    MAX(
        olympics_games[year]
    )

VAR prev_year =
        MAXX(
            FILTER(
                ALL(olympics_games[year]),
                olympics_games[year] < curr_year
            ),
            olympics_games[year]
        )

RETURN
    IF(
        ISBLANK(
            [total_events]
            ),
        BLANK(),
        CALCULATE(
            [total_events],
            ALL(
                olympics_games
             ),
            olympics_games[year] = prev_year
        )
    )
```

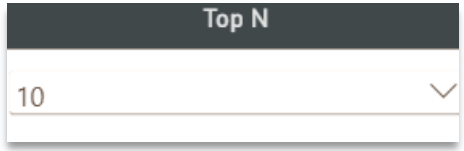
```
event_added_per_year =
            IF(
                ISBLANK([prev_year_events]),
                BLANK(),
                [total_events] - [prev_year_events]
            )
```

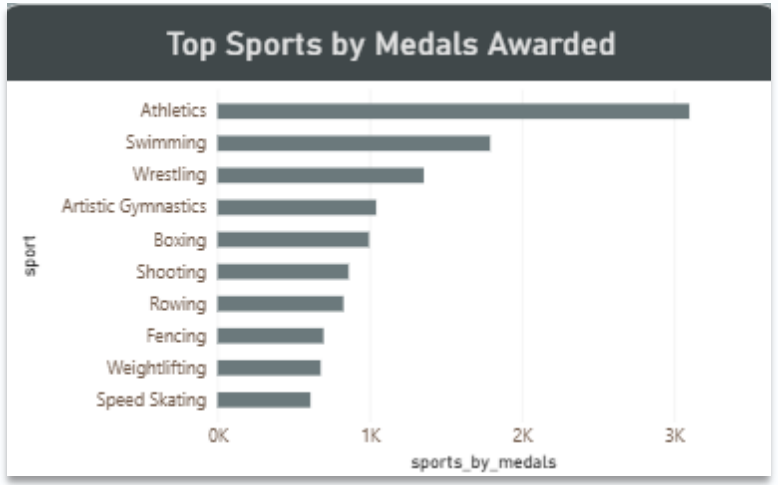



```
total_events =
    DISTINCTCOUNT(
        olympic_athlete_event_results[event]
        )
```

# Q3 Top Sports by Medals Awarded

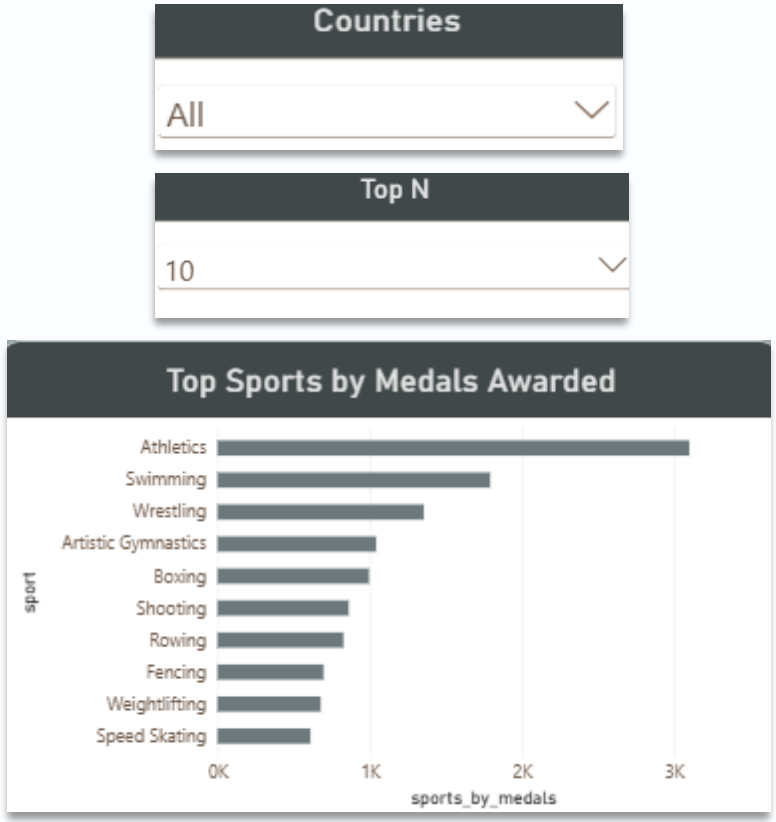
```
1  sports_by_medals =
2
3  VAR ranking =
4      RANKX(
5          ALLSELECTED(
6              olympic_athlete_event_results[sport]
7              ),
8          [total_medal_awarded], , DESC, Dense
9      )
10
11 RETURN
12     IF(
13         ranking <= Parameter[Parameter Value], [total_medal_awarded]
14     )
```



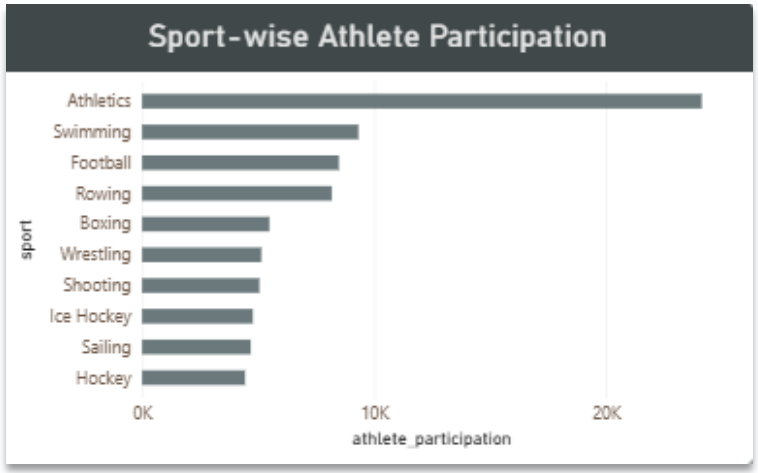
# Q4 Country Dominance in Sports

```
1  sports_by_medals =
2
3  VAR ranking =
4      RANKX(
5          ALLSELECTED(
6              olympic_athlete_event_results[sport]
7              ),
8          [total_medal_awarded], , DESC, Dense
9      )
10
11 RETURN
12     IF(
13         ranking <= Parameter[Parameter Value], [total_medal_awarded]
14     )
```



# Q5 Sport-wise Athlete Participation

```
athlete_participation =
        CALCULATE(
            DISTINCTCOUNT(
                olympic_athlete_event_results[athlete_id]
            ),
            olympic_athlete_event_results[pos] <> "DNS"
        )
```



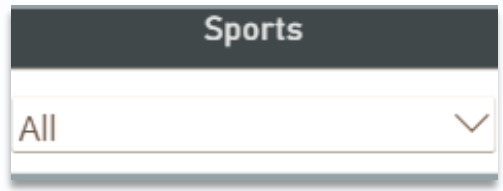
# Q6 Sport-wise Medal Trends Over Time

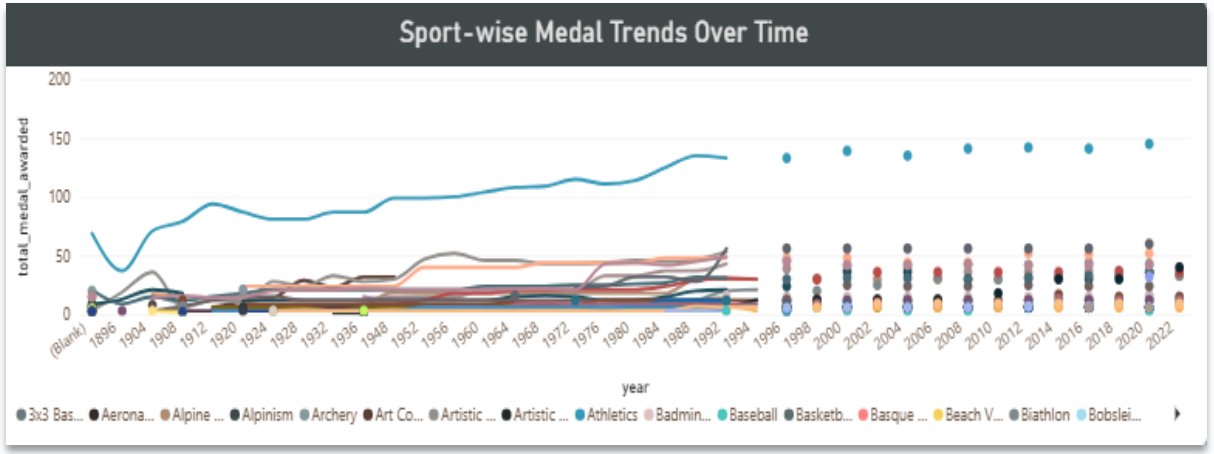
```
total_medal_awarded =

VAR IndividualMedals =
    CALCULATE(
        COUNTROWS(olympic_athlete_event_results),
        olympic_athlete_event_results[isTeamSport] = "False",
        olympic_athlete_event_results[medal] <> ""
    )

VAR TeamMedals =
    COUNTROWS(
        CALCULATETABLE(
            SUMMARIZE(
                olympic_athlete_event_results,
                olympic_athlete_event_results[result_id],
                olympic_athlete_bio[country_noc],
                olympic_athlete_event_results[medal]
            ),
            olympic_athlete_event_results[isTeamSport] = "True",
            olympic_athlete_event_results[medal] <> ""
        )
    )

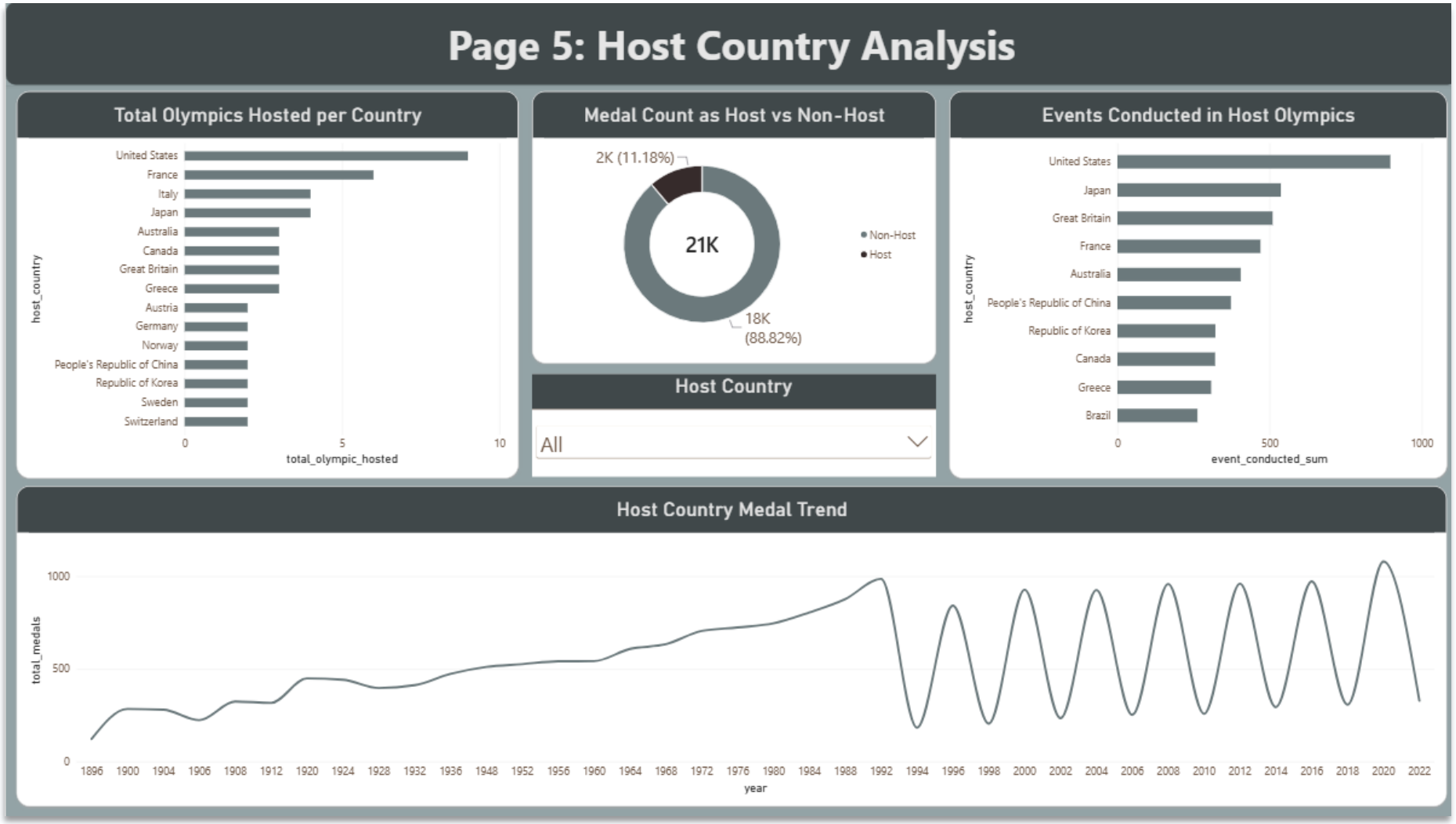
RETURN
    IndividualMedals + TeamMedals
```



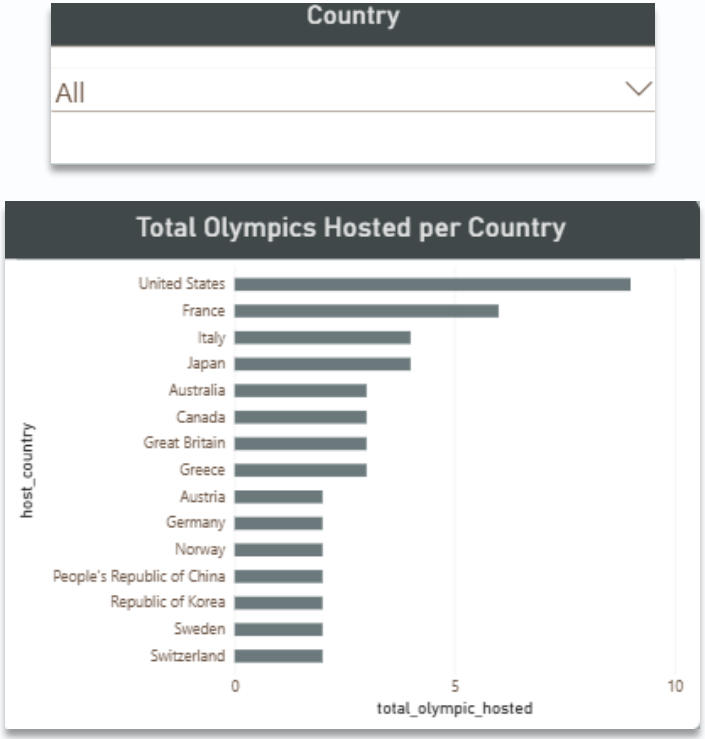
SECTION 5

# 5 Host Country Analysis



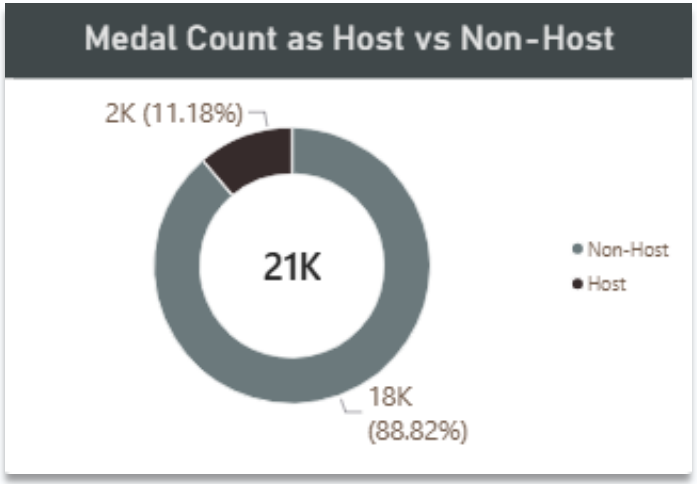
# Q1 Total Olympics Hosted per Country

```
total_olympic_hosted =
    CALCULATE(
        DISTINCTCOUNT(
            olympics_games[edition_id]
        ),
        olympics_games[isHeld] = ""
    )
```



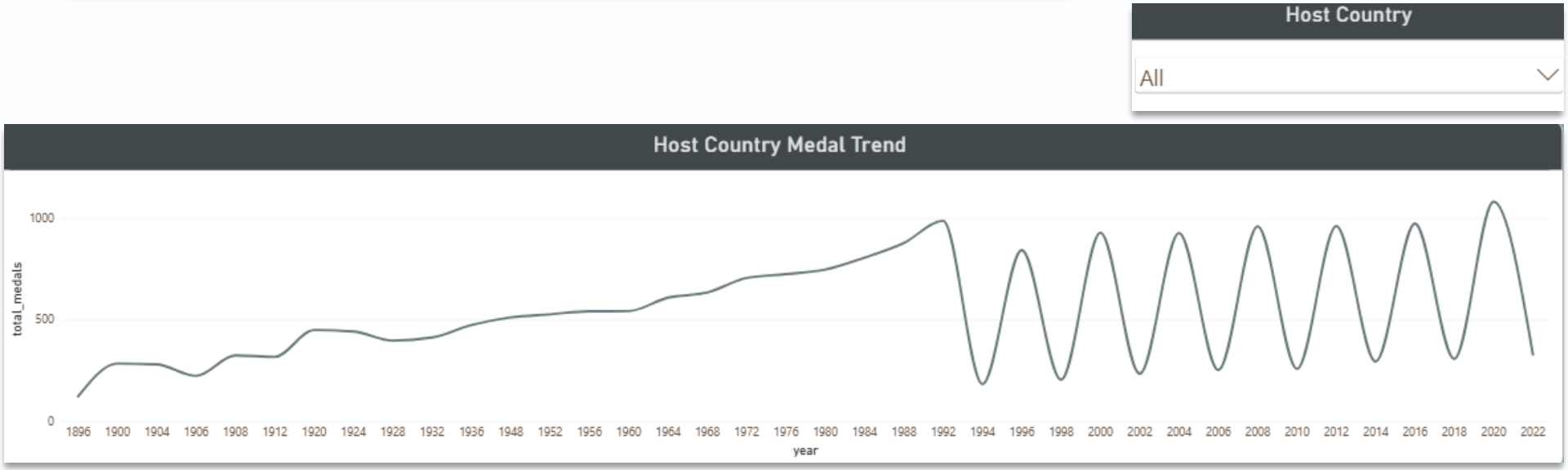
# Q2 Medal Count as Host vs Non-Host

```
1 total_medals =
2     SUMX(
3         olympic_games_medal_tally,
4         olympic_games_medal_tally[total]
5         )
```



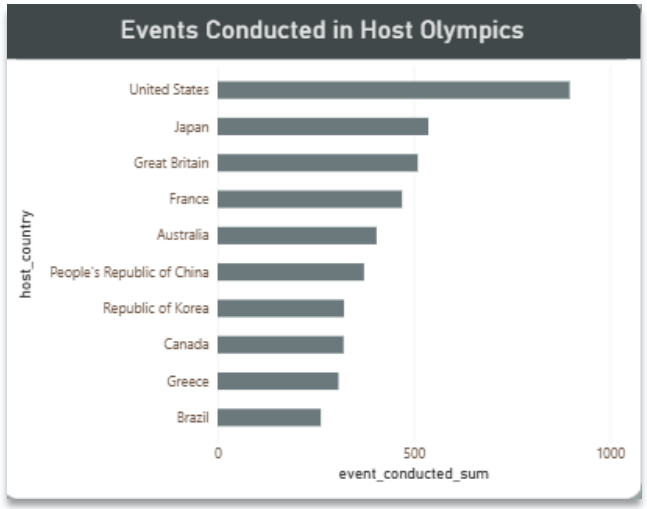
# Q3 Host Country Medal Trend

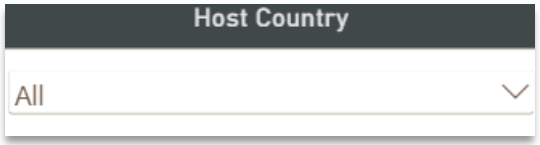
```
1 total_medals =
2     SUMX(
3         olympic_games_medal_tally,
4         olympic_games_medal_tally[total]
5         )
```



# Q4 Events Conducted in Host Olympics

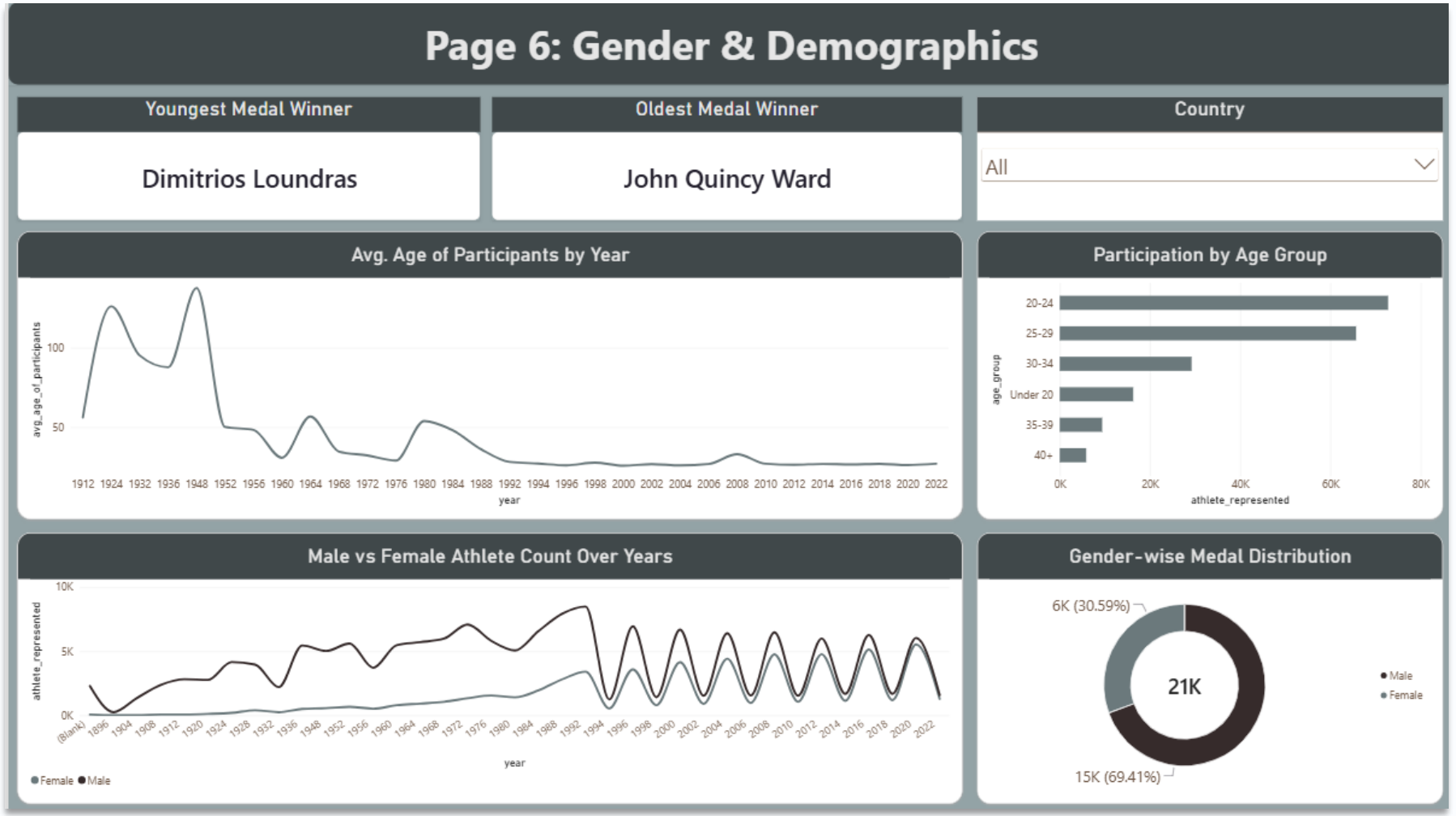
```
1 event_conducted_sum =
2         SUMX(
3             VALUES(
4                 olympics_games[edition_id]
5             ),
6             [total_events]
7         )
```



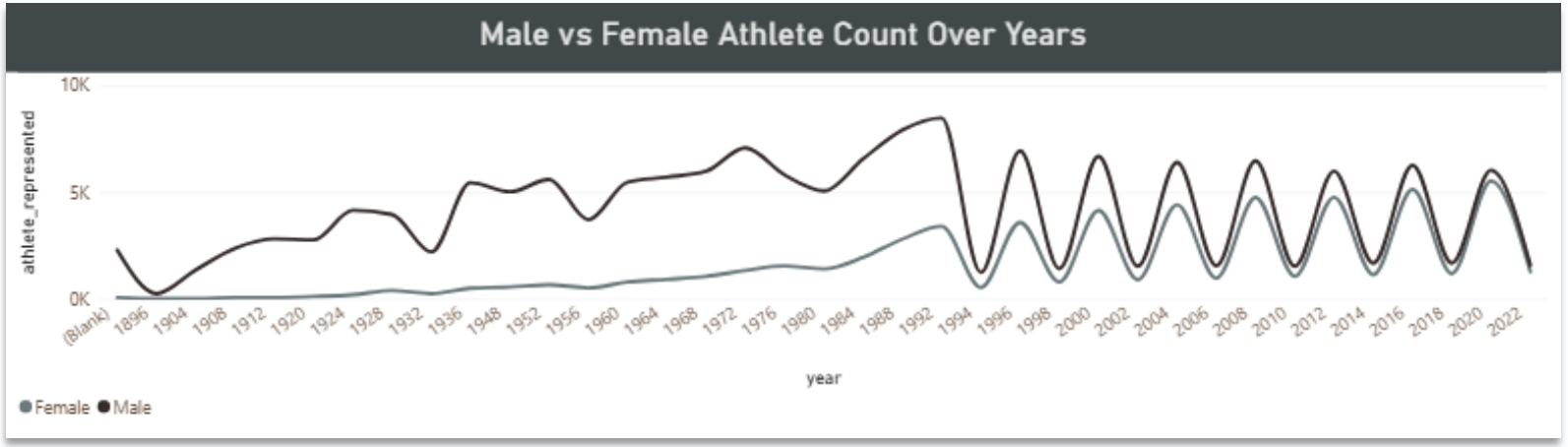
SECTION 6

# 6 Gender & Demographics



# Q1 Male vs Female Athlete Count Over Years

```
athlete_represented =
    DISTINCTCOUNT(
        olympic_athlete_event_results[athlete_id]
        )
```



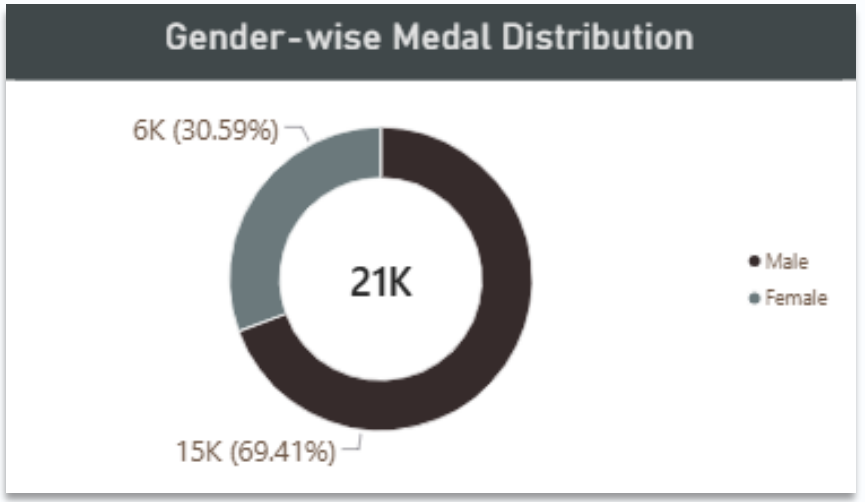
# Q2 Gender-wise Medal Distribution

```
total_medal_awarded =

VAR IndividualMedals =
    CALCULATE(
        COUNTROWS(olympic_athlete_event_results),
        olympic_athlete_event_results[isTeamSport] = "False",
        olympic_athlete_event_results[medal] <> ""
    )

VAR TeamMedals =
    COUNTROWS(
        CALCULATETABLE(
            SUMMARIZE(
                olympic_athlete_event_results,
                olympic_athlete_event_results[result_id],
                olympic_athlete_bio[country_noc],
                olympic_athlete_event_results[medal]
            ),
            olympic_athlete_event_results[isTeamSport] = "True",
            olympic_athlete_event_results[medal] <> ""
        )
    )

RETURN
    IndividualMedals + TeamMedals
```



# Q3 Youngest & Oldest Medal Winners

```
young =
    CALCULATE(
        FIRSTNONBLANK(
            olympic_athlete_event_results[athlete],
            1
        ),
        olympic_athlete_event_results[age_at_medal_wins] = MIN(olympic_athlete_event_results
        [age_at_medal_wins])
    )
```

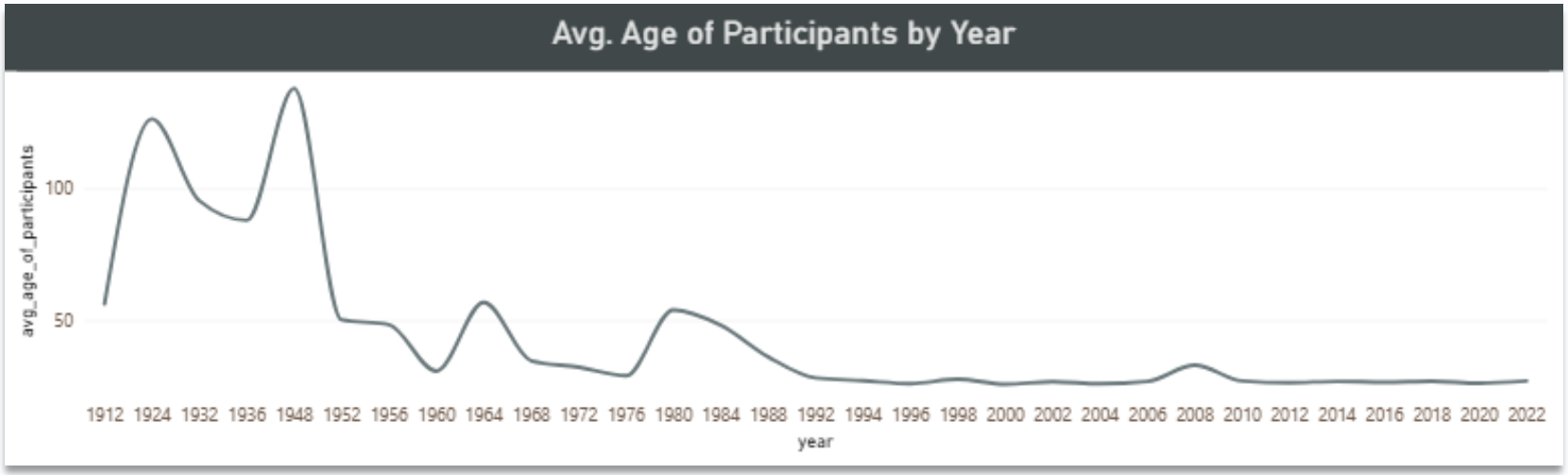
| Youngest Medal Winner |
| --- |
| Dimitrios Loundras |

```
old =
    CALCULATE(
        FIRSTNONBLANK(
            olympic_athlete_event_results[athlete],
            1
        ),
        olympic_athlete_event_results[age_at_medal_wins] = MAX(olympic_athlete_event_results
        [age_at_medal_wins])
    )
```

| Oldest Medal Winner |
| --- |
| John Quincy Ward |

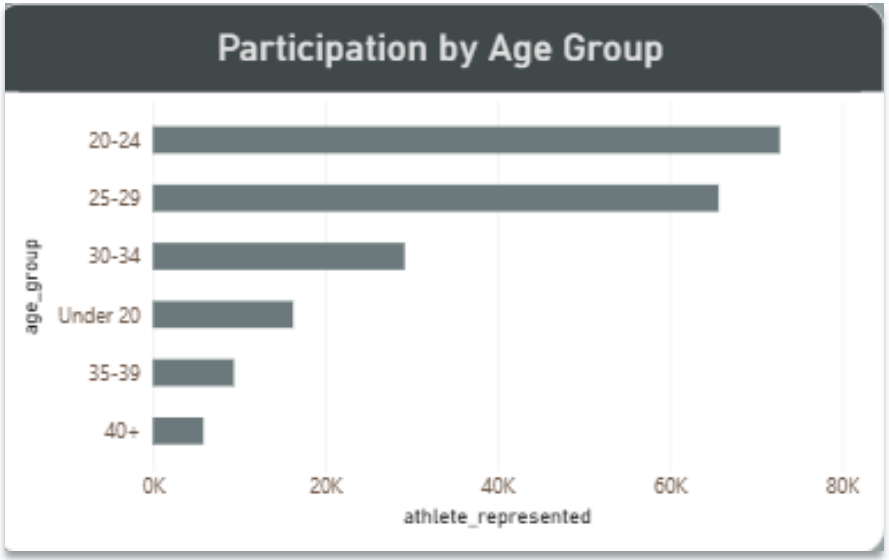
# Q4 Avg. Age of Participants by Year

```
1 avg_age_of_participants =
2     AVERAGE(
3         olympic_athlete_event_results[age_at_games]
4     )
```



# Q5 Participation by Age Group

```
athlete_represented =
    DISTINCTCOUNT(
        olympic_athlete_event_results[athlete_id]
        )
```



# Thank You

*Every great performance leaves a trail of numbers; analytics turns that trail into a medal-worthy story.*